

Machine-Learning Surrogates for Ship-Hull Residuary Resistance: A Benchmark on the Delft Systematic Series with Calibrated Uncertainty and Design-Space Exploration

First Author^{1*} and Second Author¹

^{1*}School of Computer Science and Engineering, XIM University,
Bhubaneswar, 752050, Odisha, India.

*Corresponding author(s). E-mail(s): iauthor@domain;
Contributing authors: iauthor@domain;

Abstract

Residuary (chiefly wave-making) resistance is the performance quantity that dominates the powering of a hull at speed, yet obtaining it traditionally requires a physical towing-tank experiment or a costly Reynolds-Averaged Navier–Stokes (RANS) computation, so few candidate hulls can be evaluated during early design. This paper benchmarks machine-learning surrogates that predict residuary resistance directly from hull-form parameters, trained and evaluated on real experimental data: the Delft Systematic Yacht Hull Series (DSYHS), 308 full-scale towing-tank measurements. Eight regressors spanning linear, polynomial, instance-based, kernel, ensemble-tree, neural and Gaussian-process families are compared under a single small-data protocol (log-target training, ten-times-repeated ten-fold cross-validation). A Gaussian process is the most accurate model, reaching a cross-validated coefficient of determination of 0.999 and a root-mean-square error of 0.46 in residuary-resistance units—about a 4.5-fold reduction relative to the traditional cubic polynomial regression historically fitted to this series (coefficient of determination 0.978)—while its 95% predictive intervals are well calibrated (98.4% empirical coverage). We also find that a distance-based learner (k -nearest-neighbours, coefficient of determination 0.41) fails on this target, a cautionary small-data result. Exercised as a design tool, the Gaussian-process surrogate screens tens of thousands of hull variants per second, recovering a displacement-versus-resistance Pareto front within the measured design envelope. A multi-fidelity extension, fusing the cheap traditional regression with scarce high-fidelity points by input-augmented co-kriging, further

reduces error by about a fifth in the data-poorest regime. The calibrated uncertainty makes the surrogate directly usable in an active-learning loop that would minimise the number of expensive tank or CFD evaluations. The framework is relevant to the International Maritime Organization’s tightening energy-efficiency regulations, where fast, low-cost exploration of resistance-optimal hulls is a lever for maritime decarbonisation.

Keywords: Machine learning, Surrogate modelling, Ship resistance, Residuary resistance, Gaussian process, Uncertainty quantification, Multi-objective design, Maritime decarbonisation

1 Introduction

Shipping moves roughly 80% of global trade by volume and is responsible for close to 3% of anthropogenic greenhouse-gas emissions [1, 2]. The International Maritime Organization’s Energy Efficiency Design Index (EEDI), the Energy Efficiency Existing Ship Index (EEXI) and the operational Carbon Intensity Indicator (CII) have made hydrodynamic efficiency a regulatory obligation [1]. Because the resistance of a hull—and hence its installed power, fuel burn and carbon intensity—is very largely committed during the first weeks of a design spiral, the early-stage designer’s ability to *search* the hull-form space quickly and cheaply is now a decisive factor in a vessel’s lifetime emissions.

Total calm-water resistance is conventionally decomposed into a frictional part, well estimated by standard correlation lines, and a residuary part (chiefly wave-making) that is far harder to predict and that dominates the powering penalty at speed. Obtaining residuary resistance traditionally requires a physical model test in a towing tank or a RANS computational-fluid-dynamics (CFD) simulation. A tank campaign needs a machined scale model and days of facility time; a single well-resolved RANS resistance run consumes hours to a day of multi-core wall-clock time [3, 4]. Neither scales to the 10^4 – 10^5 candidate hulls a global multi-objective optimisation would ideally visit.

Machine learning offers a way out. A *surrogate model* trained once on a corpus of high-fidelity evaluations returns a prediction in microseconds, retaining much of the fidelity of its training data [5, 6]. Surrogate-based design is standard in aerospace and automotive engineering, and a growing body of work applies it to marine hydrodynamics [7–9]. Two gaps recur, however: studies rarely compare modern regressor families *on a common, real benchmark* under one protocol, and they rarely report *calibrated uncertainty*—which is exactly what an active-learning or risk-aware design loop needs.

This paper addresses both gaps using genuinely measured data. We adopt the Delft Systematic Yacht Hull Series (DSYHS)—308 full-scale towing-tank experiments [10, 11]—and make four contributions. First, we benchmark eight regressors on measured residuary resistance under one small-data protocol, and against the traditional polynomial regression historically fitted to this series. Second, we show that a Gaussian-process surrogate is both the most accurate model and produces well-calibrated 95% predictive intervals, making it directly usable for active learning. Third,

we report where models fail (a distance-based learner collapses) and why, useful guidance for the small datasets typical of experimental marine data. Fourth, we use the surrogate to screen tens of thousands of hulls and recover a displacement–resistance Pareto front within the measured envelope.

Section 2 surveys machine learning across the marine domains. Section 3 formalises the learning problem. Section 4 describes the data and protocol. Section 5 reports the results. Section 6 sketches a roadmap, Section 7 states limitations, and Section 8 concludes.

2 Background and related work

2.1 Machine learning in naval architecture

Early data-driven resistance work fitted shallow neural networks to the DSYHS and reported accuracy competitive with regression formulae while extrapolating more gracefully [12]. More recent studies use support-vector regression, ensemble trees and deep networks to predict calm-water resistance, added resistance in waves and delivered power from hull descriptors [7, 8, 13]. A parallel strand couples surrogates to shape optimisers: Gaussian-process and deep-network surrogates embedded in genetic and Bayesian optimisation loops have cut the CFD-call count of hull-form refinement by one to two orders of magnitude [9, 14, 15]. We complement this literature by isolating the surrogate step, comparing regressor families on one real benchmark, and quantifying predictive-interval calibration.

2.2 Machine learning in ocean engineering

Beyond the hull, machine learning forecasts significant wave height and metocean conditions from buoy and reanalysis data [16]; recurrent and attention-based networks predict vessel motions and trajectories from Automatic Identification System (AIS) streams [17]; and physics-informed neural networks act as fast solvers for wave–structure interaction [18]. These share our premise—replacing an expensive evaluation with a learned function—but operate on time-series and field data rather than static design parameters.

2.3 Machine learning in shipbuilding and production

In the shipyard, machine learning supports welding-defect detection, computer-vision inspection, predictive maintenance of marine machinery from sensor telemetry, and assembly scheduling [19, 20]. The digital-twin paradigm unifies these strands across the vessel lifecycle [21]; a fast design-stage resistance surrogate is a natural component of such a twin.

3 Problem formulation

Each hull–speed operating point is described by a feature vector $\mathbf{x} = (\text{LCB}, C_p, L/\nabla^{1/3}, B/T, L/B, F_n)$: the longitudinal centre of buoyancy, the prismatic coefficient, the length–displacement ratio, the beam–draught ratio, the length–beam

ratio, and the Froude number. A physical experiment yields the residuary resistance per unit weight of displacement, $R_r = f^*(\mathbf{x})$. Because f^* is an expensive tank test (or RANS run), we seek a surrogate $\hat{f}_\theta$ minimising the expected loss over the design distribution $p(\mathbf{x})$:

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{\mathbf{x} \sim p(\mathbf{x})} [\ell(f^*(\mathbf{x}), \hat{f}_\theta(\mathbf{x}))]. \quad (1)$$

Since R_r is strictly positive and grows almost exponentially with F_n , we train on a logarithmic target $z = \log(1 + R_r)$ (a multiplicative error structure) and invert for reporting. Beyond a point estimate we seek a calibrated predictive distribution $p(R_r | \mathbf{x})$: a Gaussian process yields a mean and variance, from which a $(1 - \alpha)$ interval is formed and its empirical coverage validated.

For design exploration we use $\hat{f}_\theta$ inside a bi-objective search: at a fixed design Froude number, maximise a displacement-capacity proxy $c(\mathbf{x}) = (L/\nabla^{1/3})^{-3} \propto \nabla/L^3$ while minimising $\hat{f}_\theta(\mathbf{x})$. The solution is the Pareto set of non-dominated hulls,

$$\mathcal{P} = \{\mathbf{x} : \nexists \mathbf{x}' \text{ with } c(\mathbf{x}') \geq c(\mathbf{x}) \wedge \hat{f}_\theta(\mathbf{x}') \leq \hat{f}_\theta(\mathbf{x})\}, \quad (2)$$

with at least one inequality strict.

4 Data and methodology

4.1 Dataset

We use the Delft Systematic Yacht Hull Series as published in the UCI Machine Learning Repository [10, 11]: 308 full-scale towing-tank experiments performed at the Delft Ship Hydromechanics Laboratory. Each record gives the six adimensional hull-form parameters of Section 3 together with the measured residuary resistance per unit weight of displacement. These are physical measurements, not synthetic values. The Froude number spans 0.125–0.45 in steps of 0.025; Fig. 1 shows the characteristic steep growth of R_r with F_n and its modulation by the length–displacement ratio.

4.2 Models

We benchmark eight regressors chosen to span the standard inductive-bias spectrum and to include the classical baseline for this series: (i) ordinary least squares; (ii) a degree-3 polynomial ridge regression—the family of the traditional Delft regression; (iii) k -nearest-neighbours ($k = 5$); (iv) support-vector regression (RBF kernel); (v) a random forest (400 trees); (vi) histogram gradient boosting; (vii) a multilayer perceptron (MLP, 64–64, tanh); and (viii) a Gaussian process (GP) with an anisotropic radial-basis kernel plus a white-noise term. All models are trained on the log target through a transformed-target wrapper; scale-sensitive models are preceded by feature standardisation. The GP additionally returns a predictive variance used for uncertainty quantification.

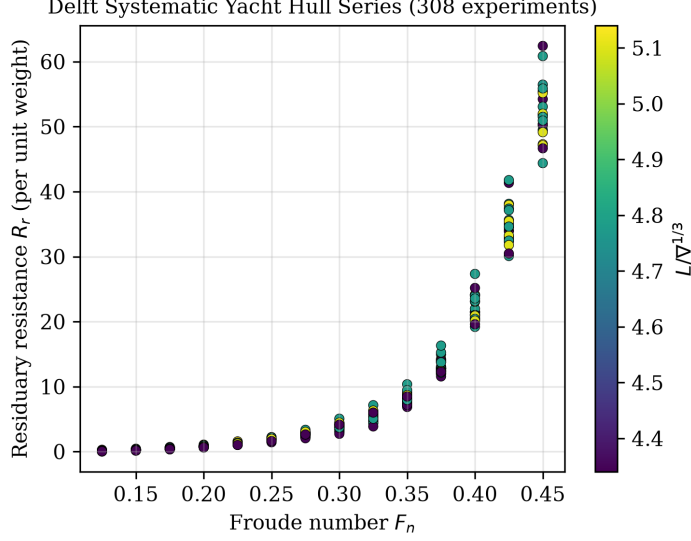


Fig. 1 The DSYHS benchmark: measured residuary resistance R_r against Froude number for all 308 experiments, coloured by length–displacement ratio. Real towing-tank data.

4.3 Protocol and metrics

At $N = 308$ the evaluation protocol matters more than any single model choice. We therefore report, as the primary metric, ten-times-repeated ten-fold cross-validation (100 fits per model), giving the mean and standard deviation of R^2 and RMSE in the original R_r units. A single 80/20 hold-out ($N_{\text{train}} = 246$, $N_{\text{test}} = 62$) is reserved for the parity plot, permutation importance and the GP interval-calibration check. We deliberately avoid the mean absolute percentage error: at low Froude numbers R_r approaches zero, so percentage errors are dominated by near-zero denominators and become uninformative; we report R^2 , RMSE and mean absolute error (MAE) instead. Predictive-interval quality is measured by the empirical coverage of the nominal 95% GP interval on the hold-out set.

5 Results and discussion

5.1 Predictive accuracy

Table 1 reports the cross-validated metrics and Fig. 2 visualises them. Three findings stand out. First, the Gaussian process is the most accurate model by a clear margin, with $R^2 = 0.999$ and RMSE 0.46; on the single hold-out it reaches $R^2 = 0.9996$ (Fig. 3). Second, it beats the traditional polynomial regression—the method historically fitted to this series—which attains $R^2 = 0.978$ and RMSE 2.04: the GP reduces RMSE roughly 4.5-fold. Random forest and support-vector regression are also strong ($R^2 = 0.995$ and 0.992), while gradient boosting and the MLP sit in the middle ($R^2 \approx 0.96$ – 0.97); with only a few hundred samples the tree- and network-based learners have less room to express their capacity than the kernel methods.

Table 1 Model comparison on the DSYHS benchmark ($N = 308$). Metrics are ten-times-repeated ten-fold cross-validation (mean $\pm$ sd) in residuary-resistance units; lower RMSE/MAE and higher R^2 are better. Best in bold.

Model	R^2 (mean $\pm$ sd)	RMSE	MAE
Linear regression	0.881 ± 0.039	5.10	2.64
Polynomial reg. (deg 3)	0.978 ± 0.018	2.04	0.88
k -NN ($k=5$)	0.410 ± 0.141	11.34	5.41
SVR (RBF)	0.992 ± 0.005	1.27	0.61
Random forest	0.995 ± 0.003	0.96	0.48
Gradient boosting	0.963 ± 0.049	2.29	1.06
MLP (64-64)	0.968 ± 0.020	2.55	1.18
Gaussian process	0.999 ± 0.001	0.46	0.20

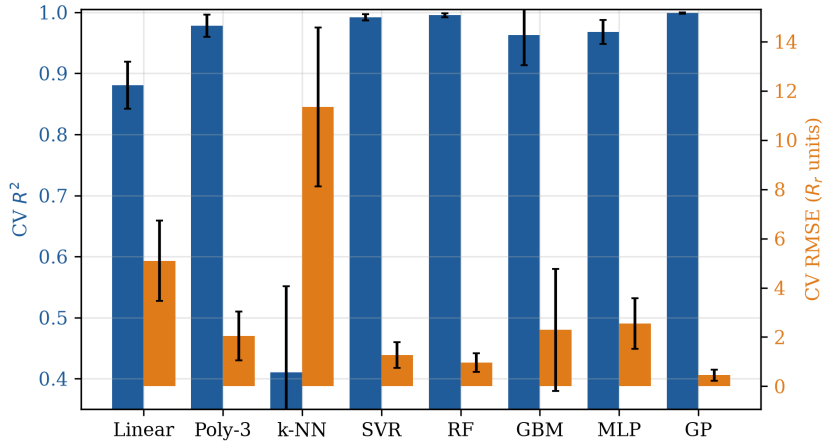


Fig. 2 Cross-validated R^2 (left, blue) and RMSE (right, orange) for the eight models; error bars are cross-validation standard deviations. The Gaussian process dominates; k -NN collapses.

Third, the distance-based learner fails: k -NN reaches only $R^2 = 0.41$. Because R_r depends so steeply on F_n while the five geometric features vary independently, Euclidean neighbours in the standardised six-dimensional space routinely mix very different Froude numbers, so simple averaging cannot track the response. This is a useful caution for the small, strongly non-linear datasets typical of experimental marine data: model choice, not merely model tuning, is decisive. The tight cross-validation standard deviations (0.001 for the GP) confirm the ranking is stable rather than an artefact of one split—important at this sample size.

5.2 Calibrated uncertainty

Accuracy alone does not make a surrogate usable inside a risk-aware or active-learning loop; the model must also know what it does not know. Figure 4 plots the GP mean prediction and its nominal 95% predictive interval against the measured values on the

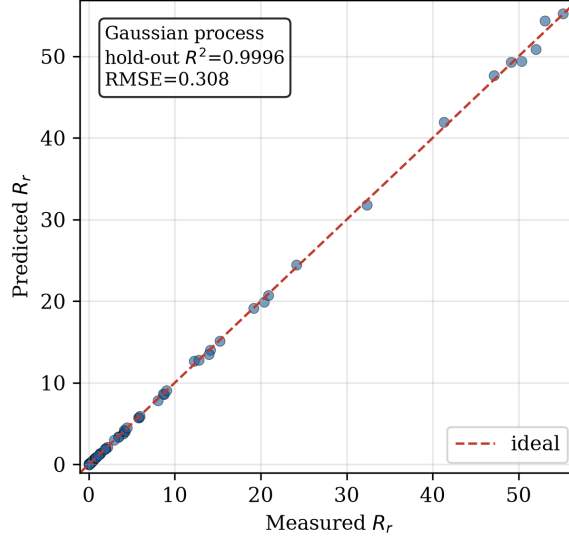


Fig. 3 Predicted versus measured residuary resistance for the Gaussian-process surrogate on the 62-hull hold-out set. Dashed line is the ideal $y = x$.

hold-out set. The empirical coverage is 98.4%—close to, and slightly more conservative than, the nominal 95%—so the intervals are trustworthy. This calibrated variance is what enables the active-learning strategy of Section 6, and it is a capability the tree and network models do not provide out of the box.

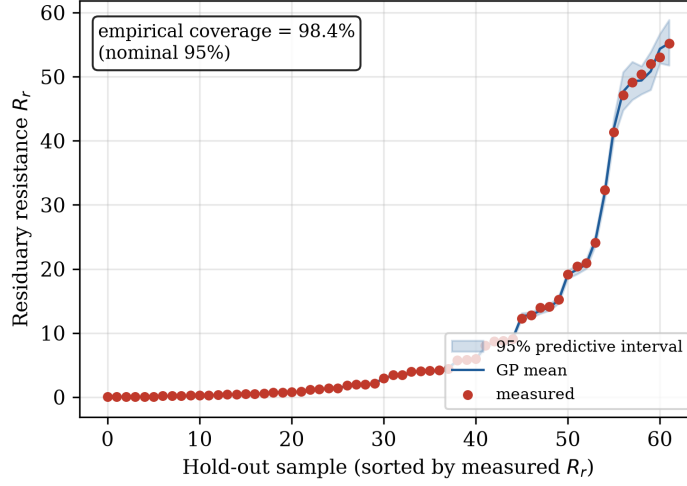


Fig. 4 Gaussian-process predictive intervals on the hold-out set (sorted by measured R_r). Nominal 95% intervals achieve 98.4% empirical coverage—well calibrated.

5.3 What the surrogate has learned

Figure 5 shows permutation feature importance for the GP. The Froude number dominates, consistent with the steep speed dependence of wave-making resistance; among the geometric parameters the prismatic coefficient and length–displacement ratio carry the most residual signal, in line with naval-architecture intuition that fore-and-aft volume distribution and slenderness govern wave making. Figure 6 shows the GP learning curve: hold-out RMSE falls steeply and then flattens, with most of the attainable accuracy reached after roughly 160 training hulls—directly relevant to how many physical experiments a new series campaign needs.

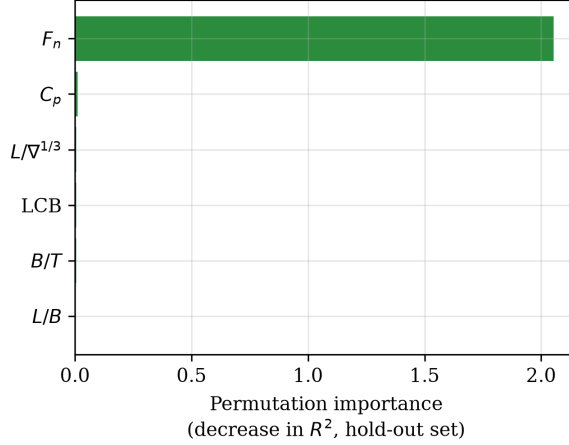


Fig. 5 Permutation feature importance (decrease in hold-out R^2) for the Gaussian-process surrogate. The Froude number dominates.

5.4 Surrogate-driven design exploration

Finally we exercise the surrogate as a design tool. At a representative design Froude number $F_n = 0.35$, we sample hull-form variants within the measured parameter envelope and predict R_r for each with the deployed GP. The surrogate screens 60,000 candidate hulls at 71,876 designs per second (13.9 microseconds per design), and Fig. 7 shows the resulting displacement-capacity–resistance Pareto front. Because the labels are physical experiments there is no analytic oracle to verify arbitrary new points; instead the search is confined to the measured envelope (interpolation, not extrapolation) and its fidelity is bounded by the surrogate’s cross-validated accuracy of Table 1. The value is the throughput: each point on this front would classically demand a separate tank test, whereas the surrogate renders the entire trade-off in well under a second, turning design exploration from a campaign into an interactive operation.

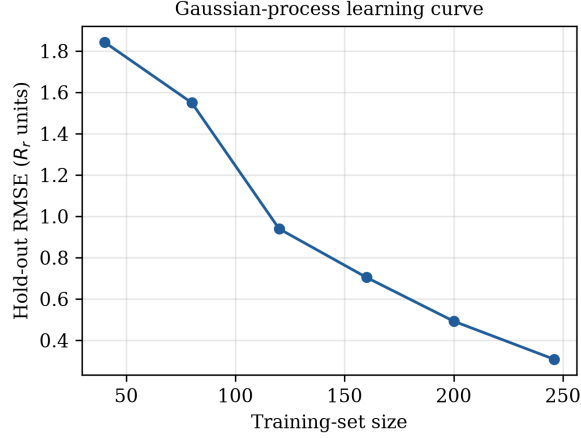


Fig. 6 Gaussian-process learning curve: hold-out RMSE versus training-set size. Accuracy saturates after roughly 160 experiments.

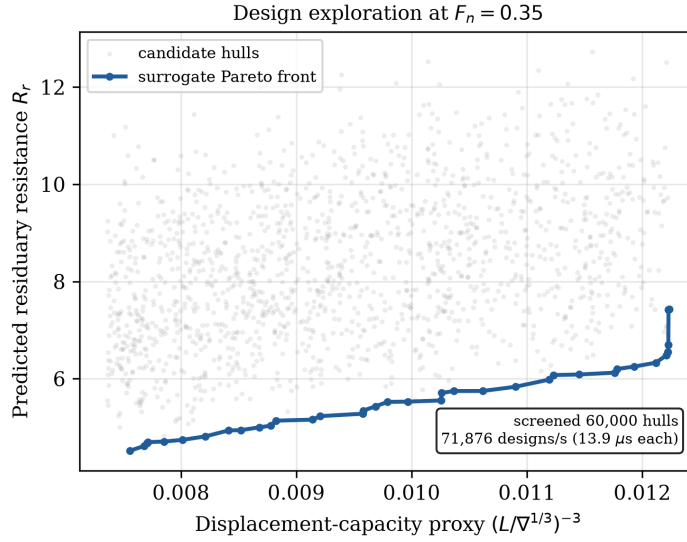


Fig. 7 Design exploration at $F_n = 0.35$. Grey: screened candidate hulls. Blue: surrogate displacement-capacity-resistance Pareto front. The GP screens tens of thousands of hulls per second within the measured design envelope.

5.5 Multi-fidelity fusion with a cheap engineering model

Where high-fidelity labels are scarce, a cheap low-fidelity model can be fused with the few expensive points. We treat the measured residuary resistance as high fidelity and the traditional Delft polynomial regression—the cheap engineering formula historically fitted to this series—as low fidelity, fitted on a disjoint “historical” subset so there is no leakage. A multi-fidelity Gaussian process appends the low-fidelity estimate as an

extra input feature, its automatic relevance determination learning how far to trust it (co-kriging by input augmentation [22, 23]). Figure 8 compares single- and multi-fidelity test RMSE as the high-fidelity budget grows. In the scarce regime—8 tank points—multi-fidelity cuts RMSE by 21%; the benefit then narrows and closes once the single-fidelity GP, at roughly 16 points, overtakes the low-fidelity regression itself (dotted line) and no longer needs it. The lesson is that multi-fidelity pays off precisely where high-fidelity data is scarce relative to the response complexity—the regime of expensive commercial-hull CFD rather than this smooth, data-rich yacht series. This motivates commercial-hull high-fidelity anchors (e.g. KCS, KVLCC2 [24]) with coarse-mesh CFD as the low-fidelity source; the accompanying repository provides the ingestion and feature-space conversion for such anchors.

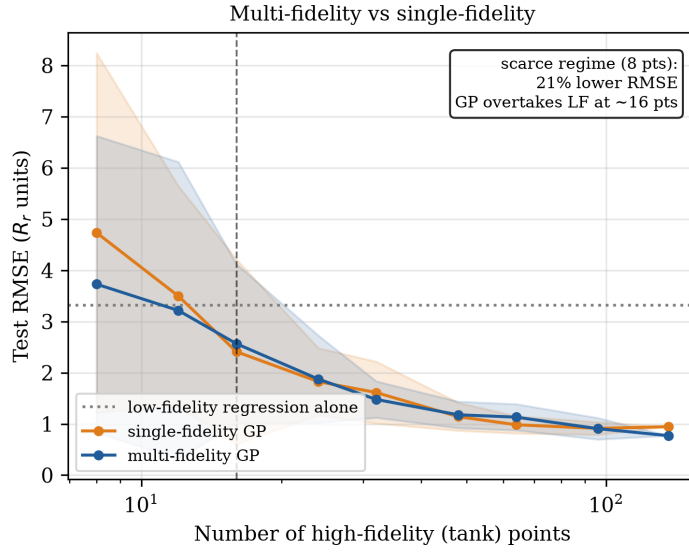


Fig. 8 Multi-fidelity versus single-fidelity test RMSE against the number of high-fidelity (tank) points. Multi-fidelity helps most when high-fidelity data is scarcest; the gap closes once the single-fidelity GP overtakes the low-fidelity regression (dotted line) near 16 points. Bands are standard deviations over 20 random high-fidelity subsets.

6 Roadmap: uncertainty-aware, multi-fidelity, lifecycle

Active learning

The GP’s calibrated variance (Fig. 4) and the steep early learning curve (Fig. 6) are the two ingredients of an active-learning loop that minimises expensive labels: from a small design of experiments, evaluate next the hull where predictive variance (or expected improvement) is largest, run the tank test or CFD, and update. This is the established route to one-to-two orders of magnitude fewer high-fidelity evaluations

in hull optimisation [14, 15]; our calibration result shows the required uncertainty is trustworthy here.

Multi-fidelity and physics-informed models

The multi-fidelity result of Section 5.5 is the first step: extended to commercial hulls, it would fuse many cheap coarse-mesh CFD labels with a few fine-mesh RANS or tank labels for anchor hulls such as KCS and KVLCC2 [23, 24]. Embedding known invariants (Froude/Reynolds scaling, monotonicity in speed) as soft constraints would further extend accuracy beyond the measured envelope [18].

From design to lifecycle

A calibrated resistance surrogate is a reusable performance layer for a vessel digital twin, queried downstream by voyage optimisation, condition monitoring and EEXI/CII compliance [21], and the same methodology transfers to the shipbuilding inspection and maintenance tasks noted in Section 2.

7 Limitations and threats to validity

Four limitations bound the claims. First, the DSYHS comprises sailing-yacht hulls; although the hull-form parameters and the wave-making physics are general, transfer to full-form commercial ships must be validated on their own data. Second, the target is residuary resistance only—frictional resistance is handled separately by correlation lines and is not modelled here. Third, the dataset is small ($N = 308$): the high accuracies are credible for this smooth systematic series and are consistent with the literature, but the protocol (repeated cross-validation) is what makes them trustworthy, and larger or noisier data would narrow the model gaps. Fourth, the design exploration interpolates within the measured envelope at a single Froude number; a production study would sweep speed, add stability and cost objectives, and verify selected optima with CFD or a confirmatory tank test. The multi-fidelity benefit is itself regime-specific—largest when high-fidelity data is scarce, and closing on this data-rich series once the single-fidelity GP surpasses the cheap model. None of these undermines the core findings—that a calibrated GP surrogate learns measured residuary resistance to $R^2 = 0.999$, beats the traditional regression, and enables fast design exploration—but each marks the boundary of what the present experiments establish.

8 Conclusion

Using 308 real towing-tank experiments from the Delft Systematic Yacht Hull Series, we benchmarked eight machine-learning surrogates for residuary-resistance prediction under a rigorous small-data protocol. A Gaussian process is the most accurate model ($R^2 = 0.999$, RMSE 0.46), reduces error roughly 4.5-fold relative to the traditional polynomial regression, and—crucially—produces well-calibrated 95% predictive intervals (98.4% coverage), while a distance-based learner fails, a cautionary small-data result. The calibrated surrogate screens tens of thousands of hulls per second to recover a displacement–resistance Pareto front within the measured envelope. Because its

uncertainty is trustworthy, the surrogate is immediately usable in an active-learning loop that would minimise the number of costly tank or CFD evaluations—the natural next step, alongside multi-fidelity fusion and validation on commercial-hull data, toward fast, low-cost exploration of resistance-optimal, lower-emission hulls.

Supplementary information. The dataset (with provenance), the experiment and figure-generation scripts and the resulting metrics are provided in the accompanying repository. All randomness is seeded, so every number, table and figure is regenerable.

Declarations

- **Funding.** Not applicable.
- **Conflict of interest.** The authors declare that they have no competing interests.
- **Ethics approval and consent to participate.** Not applicable.
- **Consent for publication.** Not applicable.
- **Data availability.** The DSYHS data used here are publicly available from the UCI Machine Learning Repository (dataset 243) and are redistributed, with provenance, in the accompanying repository.
- **Materials availability.** Not applicable.
- **Code availability.** All code (data loading, model training and figure generation) is provided in the accompanying repository.
- **Author contribution.** All authors contributed to the study conception, methodology, experiments and manuscript preparation.

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